

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1715

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(*BL3*)

Assessment Tool Weightage: 20%

QUESTIONS: 3

Duration : 45 Minutes

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _K.Yamini(192572181)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K.Yamini(192572181)

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i.D1 cannot work Night shift
- ii.D2 must work before D3 (Morning < Afternoon < Night)
- iii.Only one doctor per shift

iv.D3 cannot work Morning

v.All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2.A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G).

Obstacles block certain paths.

Constraints:

i.Movement allowed: Up, Down, Left, Right

ii.Cost of each move = 1

iii.Cannot pass through obstacles

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3.An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality

- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Total Marks (30)
Problem Understanding	25% (2.5)	/2.5	/2.5	/2.5	/7.5
Constraint Analysis	25% (2.5)	/2.5	/2.5	/2.5	/7.5
Solution Development	25% (2.5)	/2.5	/2.5	/2.5	/7.5
Application of Concepts	15% (1.5)	/1.5	/1.5	/1.5	/4.5
Justification	10% (1)	/1	/1	/1	/3
Total per Question	10 Marks	/10	/10	/10	/30

1.Backtracking Search for Doctor Shift Assignment

1.Problem Understanding

The problem is to assign three doctors D1, D2, and D3 to three shifts Morning, Afternoon, and Night. Each doctor must be assigned exactly one shift and each shift must have exactly one doctor. The assignment must satisfy all given constraints. This is a Constraint Satisfaction Problem where we must find a consistent assignment.

2.Constraint Analysis

The constraints are:

D1 cannot work the Night shift, so D1 can only take Morning or Afternoon.

D2 must be scheduled before D3 in the order Morning < Afternoon < Night, meaning the shift assigned to D2 must come earlier than the shift assigned to D3.

Only one doctor can be assigned to each shift, so no two doctors can share the same shift.

D3 cannot work the Morning shift, so D3 can take only Afternoon or Night.

All doctors must be assigned exactly one shift, so no doctor is left unassigned.

These constraints reduce the possible combinations and guide the search process.

3.Solution Development using Backtracking

Step 1: Start with an empty assignment

Assignment = {}

Step 2: Assign D1

Try Morning

Check constraint: D1 is allowed Morning

Assignment = {D1 = Morning}

Step 3: Assign D2

Try Morning

Rejected because Morning is already assigned

Try Afternoon

Valid assignment

Assignment = {D1 = Morning, D2 = Afternoon}

Step 4: Assign D3

Try Morning

Rejected because D3 cannot work Morning

Try Afternoon

Rejected because already assigned

Try Night

Check order constraint

D2 = Afternoon and D3 = Night, so Afternoon < Night is satisfied

Assignment = {D1 = Morning, D2 = Afternoon, D3 = Night}

Step 5: Final Constraint Check

Each doctor has one shift

Each shift has one doctor

All constraints are satisfied

4.Application of Concepts

Backtracking is used to solve constraint satisfaction problems by assigning values step by step and checking constraints at each stage. If a constraint is violated, the algorithm backtracks and tries another option. This avoids unnecessary exploration of invalid solutions.

5.Justification

Backtracking is suitable because it systematically explores possible assignments and eliminates invalid ones early. It ensures that a valid solution is found without checking all combinations.

Final Answer

D1 is assigned Morning

D2 is assigned Afternoon

D3 is assigned Night

2. Breadth First Search in 5×5 Grid

1.Problem Understanding

A robot must move from Start position S to Goal position G in a 5×5 grid. The robot can move up, down, left, and right. Some cells contain obstacles and cannot be crossed. Each movement has equal cost of 1. The objective is to find the shortest path using Breadth First Search.

2.Constraint Analysis

Movement is restricted to four directions only.

Obstacles block certain cells and must be avoided.

Each step has equal cost, so the shortest path is the path with minimum number of steps.

The robot must not revisit already visited nodes to avoid loops. These constraints make BFS an appropriate method.

3.Solution Development using BFS

Grid Representation

S 0 0 X 0

0 X 0 X 0

0 X 0 0 0

0 0 X X 0

X 0 0 G 0

Step 1: Initialize

Queue = [S]

Visited = {S}

Step 2: Expand S

Add (0,1) and (1,0)

Queue = [(0,1), (1,0)]

Step 3: Expand (0,1)

Add (0,2)

Queue = [(1,0), (0,2)]

Step 4: Expand (1,0)

Add (2,0)

Queue = [(0,2), (2,0)]

Step 5: Expand (0,2)

No valid move due to obstacle

Queue = [(2,0)]

Step 6: Expand (2,0)

Add (3,0)

Queue = [(3,0)]

Step 7: Expand (3,0)

Add (3,1)

Queue = [(3,1)]

Step 8: Expand (3,1)

Add (4,1)

Queue = [(4,1)]

Step 9: Expand (4,1)

Add (4,2)

Queue = [(4,2)]

Step 10: Expand (4,2)

Add Goal node G

Queue = [G]

Step 11: Goal reached

Stop search

4.Application of Concepts

BFS is an uninformed search algorithm that explores nodes level by level. It uses a queue to maintain the order of expansion. BFS guarantees the shortest path when all edge costs are equal.

5.Justification

BFS is the best choice because all moves have equal cost. It ensures that the first time the goal is reached is through the shortest path.

Optimal Path

S to (1,0) to (2,0) to (3,0) to (3,1) to (4,1) to (4,2) to G

Cost

Total steps = 7

Total cost = 7

3. Autonomous Rescue Robot Problem

1.Problem Understanding

The robot operates in a grid environment where it must move from start position S to goal position G. The environment is partially observable, meaning the robot cannot see the entire grid. Some cells are blocked and some are risky with additional cost. The robot must find a path with minimum total cost while updating its knowledge dynamically.

2.Constraint Analysis

The robot can move only in four directions.

Obstacles block movement and must be avoided.

The robot can observe only adjacent cells, so it does not know the full environment initially.

Each normal move has cost 1 and risky cells add an extra cost of 2.

The robot must minimize total cost.

The robot must update its knowledge while moving.

These constraints require adaptive and cost-based decision making.

3.Solution Development

The suitable approach is Uniform Cost Search combined with Online Search.

Steps

Step 1: Start at position S

Step 2: Use a priority queue ordered by path cost

Step 3: Insert start node with cost 0

Step 4: Expand the node with the lowest cost

Step 5: For each adjacent cell

If it is normal, cost = previous cost + 1

If it is risky, cost = previous cost + 3

Step 6: Add new nodes to the priority queue

Step 7: Update knowledge of environment after each move

Step 8: Continue until goal G is reached

4.Application of Concepts

Intelligent Agent

The robot acts as an intelligent agent that perceives its environment and takes actions to achieve its goal.

Rationality

The robot behaves rationally by choosing actions that minimize total cost and avoid unnecessary risk.

Environment Type

The environment is partially observable because the robot cannot see the full grid.

It is dynamic because the robot discovers new information during movement.

It is discrete because it consists of grid cells.

5.Justification

Uniform Cost Search is appropriate because it handles different costs and guarantees an optimal solution. Online search is necessary because the robot does not know the environment completely at the start. This combination ensures safe and efficient navigation.

Conclusion

The robot successfully reaches the goal by minimizing cost and adapting to the environment dynamically, demonstrating effective use of AI concepts.